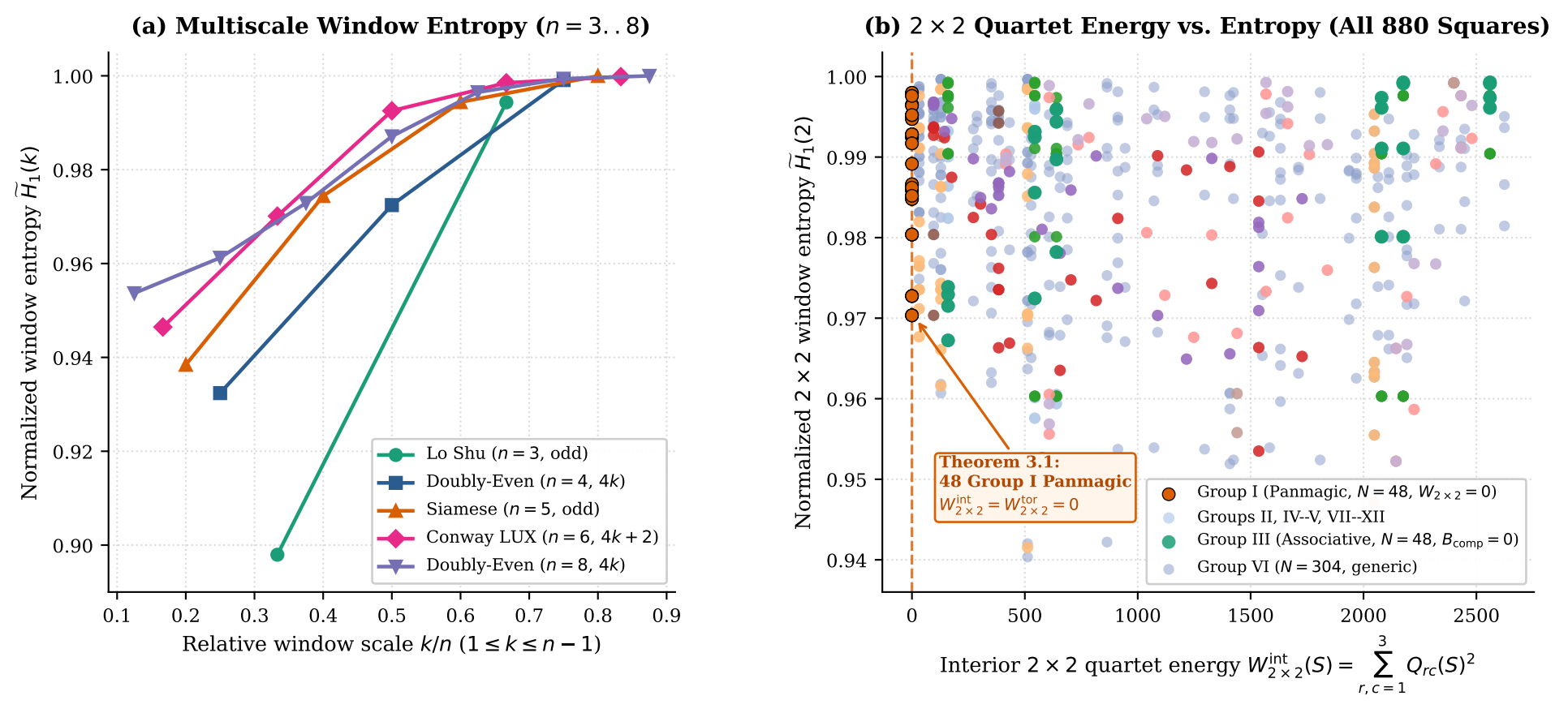

(a) Multiscale Window Entropy ($n = 3..8$)
Normalized window entropy $\widetilde{H}_1(k)$
Relative window scale $k/n$ $(1 \le k \le n-1)$
Lo Shu ($n = 3$, odd)
Doubly-Even ($n = 4$, $4k$)
Siamese ($n = 5$, odd)
Conway LUX ($n = 6$, $4k+2$)
Doubly-Even ($n = 8$, $4k$)
(b) $2 \times 2$ Quartet Energy vs. Entropy (All 880 Squares)
Normalized $2 \times 2$ window entropy $\widetilde{H}_1(2)$
Interior $2 \times 2$ quartet energy $W^{\text{int}}_{2\times 2}(S) = \sum_{r,c=1}^{3} Q_{rc}(S)^2$
Theorem 3.1:
48 Group I Panmagic
$W^{\text{int}}_{2\times 2} = W^{\text{tor}}_{2\times 2} = 0$
Group I (Panmagic, $N = 48$, $W_{2\times 2} = 0$)
Groups II, IV--V, VII--XII
Group III (Associative, $N = 48$, $B_{\text{comp}} = 0$)
Group VI ($N = 304$, generic)